

Semyon Fox

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Programmer, swimmer, filmmaker.

Third-year Computer Science and IT student at the University of Galway. I build software for study and swimming, write tools to simplify everyday tasks, and run my own Linux infrastructure. I use AI throughout development and take responsibility for reviewing, testing and maintaining what I build.

Education

University of Galway

Expected August 2028

BSc (Honours) Computer Science and Information Technology

Third year; current overall average: 2:1.

Coursework: Software Engineering; Data Structures and Algorithms; Database Systems; Networks and Data Communications; Human-Computer Interaction.

Technical skills

Languages: TypeScript, Python, Go, Rust, SQL, Bash; Java and C coursework.

Applications and data: React, Next.js, Node.js, PostgreSQL, Redis, Qdrant, REST, MCP.

Delivery: Git, Linux, Docker, Jenkins, BuildKit, Nginx, Cloudflare; AWS project experience.

Selected projects

OghmaNotes – AI study platform

Deployed

Next.js, TypeScript, PostgreSQL, Qdrant, Redis/BullMQ, Docker

oghmanotes.ie

- Built in a three-person team to bring Canvas course material, notes, cited AI chat, quizzes and spaced repetition into one study workspace.
- Developed asynchronous imports with PDF extraction, embeddings, BullMQ workers, live job status, retries and cancellation; used Qdrant for retrieval and PostgreSQL for content and ownership.
- Deployed on AWS, then moved the application and workers to self-hosted Docker to control costs.

Uisce – swim club management

Deployed

React, Express, PostgreSQL, Redis, JSON:API, Docker, Vitest

swim.semyon.ie

- Built a swimming-club app to manage squads, training sessions, attendance and results, with performance tracking.
- Designed the React interface, JSON:API backend and multi-schema PostgreSQL model; used Vitest for client and API tests, including 90+ backend tests for workflows and data access.

Home lab and CI/CD infrastructure

Self-hosted

Linux, Docker, Jenkins, Nginx, Cloudflare, Btrfs, NFS

- Run 30+ services in 54 containers, with VPN access, Cloudflare tunnels and Nginx reverse proxies across 21 internal virtual hosts.
- Tuned Jenkins pipelines across six applications with persistent BuildKit caching; cut the Irish Rail project's build from 9m 43s to 1m 11s.
- Back up services to a RAID NAS over NFS using Btrfs snapshots and daily, weekly, monthly, and yearly retention.

Irish Rail data pipeline

Deployed

Python asyncio/aiohttp, TimescaleDB, Rust axum, Docker

traein.semyon.ie

- Built a Python service polling the Irish Rail API every three seconds and storing train positions and station data in TimescaleDB.
- Containerised the pipeline and built a Rust API for a live map and delay dashboard.

Experience

School IT support Coláiste an Eachréidh	2023–2026 One voluntary, then two paid
▪ Configured laptops, set up ebooks and carried out repairs for students. ▪ Built API integrations and automation tools for ebook setup at the school.	
Laptop repair technician Cahill Computers, Athenry	2022 Part-time
▪ Diagnosed faults, replaced screens and batteries, upgraded RAM and storage, cloned drives and installed Windows.	
Laptop repair placement Lapteck, Galway	2022 Transition Year
▪ Completed a work-experience placement learning laptop diagnostics and repair.	
Kitchen porter Old Barracks, Athenry	Seasonal and part-time
▪ Supported the kitchen team during busy summer services, with additional occasional part-time work.	

Leadership

University of Galway Computer Society Treasurer since March 2026; previously Auditor and PR lead	Nov 2024 to present
▪ Organised CTF 2026 as part of the committee: 110 participants, four corporate sponsors and 50% lower participant costs. ▪ Contributed to the society website, including CI build checks, deployment and routing fixes, and committee updates.	

Open source and tools

Noctalia – Bluetooth fix C++	Merged contribution Pull request #4145
▪ Contributed a fix that stops Bluetooth discovery after ten seconds and cancels scans when the panel closes.	
Seol – temporary page publishing Go, SQLite, Docker	Open source github.com/semyonfox/seol
▪ Created a service and CLI that turn HTML reports or static sites into temporary shareable links. ▪ Added archive path and size checks, automatic expiry, rate limits and atomic replacement so a page can be updated at the same URL.	
Canvas MCP – API integration TypeScript, Model Context Protocol SDK, Zod	Open source github.com/semyonfox/canvas-mcp
▪ Consolidated and extended community tool designs into a typed MCP server exposing selected Canvas LMS APIs to assistants. ▪ Added validated inputs, pagination, request timeouts, and bounded retries for read operations.	

Awards

CompSoc: Best Intervarsity at University of Galway, 2025 and 2026; BICS National Society Award, winner 2025 and nominee 2026.

Brian Ó Maoilchiaráin Award: Outstanding Leaving Certificate student, Coláiste an Eachréidh, June 2024.

GRETB STEM Award, June 2024.

Interests

Competitive swimming and regular meets; video production, colour grading and VFX in DaVinci Resolve; woodworking and hand-built live-edge furniture.